Mark schemes

Q1.

(a) $\int xy\,\mathrm{d}x = \int_0^1 kx^{1.5}\,\mathrm{d}x$

Use of $\int xy\,dx$ or full first principles approach using strips and moments leading to appropriate integral

M1

$= \left[\dfrac{kx^{2.5}}{2.5}\right]_0^1$

$= \dfrac{k}{2.5} = \dfrac{2k}{5}$

Correct integration and substitution of limits

A1

$\Rightarrow \bar{x} = \dfrac{\int xy\,\mathrm{d}x}{\int y\,\mathrm{d}x} = \dfrac{\frac{2k}{5}}{A} = \dfrac{2k}{5A}$

Printed answer – fully correct justification required

A1

3

(b) $\int \frac{1}{2}y^2\,\mathrm{d}x = \int_0^1 \dfrac{k^2x}{2}\,\mathrm{d}x$

Use of $\int \frac{1}{2}y^2\,dx$ or $\int y^2\,dx$
Allow full first principles approach using strips and moments leading to appropriate integral

M1

$= \left[\dfrac{k^2x^2}{4}\right]_0^1$

Correct integration using fully correct integral

$= \dfrac{k^2}{4}$

A1

$\Rightarrow \bar{y} = \dfrac{k^2}{4A}$

Correct substitution of limits and division by area to *obtain correct answer*

A1

3

(c) On line $y = x \Rightarrow \bar{x} = \bar{y}$

$\frac{2k}{5A} = \frac{k^2}{4A}$

Use of $\bar{x} = \bar{y}$

M1

$k = \frac{8}{5}$

CAO

A1

2

[8]

Q2.

(a) Area = $\int_0^2 kx^3 \, dx = \left[\frac{kx^4}{4}\right]_0^2$

Attempt to integrate

M1

$= 4k$

A1

2

(b) $\int xy \, dx = \int_0^2 kx^4 dx = \left[\frac{kx^5}{5}\right]_0^2$

Attempt to use $\int xy \, dx$

M1

$= \frac{32k}{5}$

A1

$\bar{x} = \frac{\int xy \, dx}{\int y \, dx} = \frac{\frac{32k}{5}}{4k}$

Forming equation to find $\bar{x}$

M1

= 1.6

ft 'their' part (a); must not contain k

A1F

4

(c) (i) $\frac{1}{2}\int y^2\,dx = \frac{1}{2}\int_0^2 k^2 x^6\,dx = \left[\frac{k^2 x^7}{14}\right]_0^2$

Attempt to use $\frac{1}{2}\int y^2\,dx$

M1

$= \frac{64}{7}k$

A1

$\bar{y} = \frac{\frac{1}{2}\int y^2\,dx}{\int y\,dx} = \frac{\frac{64k^2}{7}}{4k} = \frac{16k}{7}$

Finding $\bar{y}$ *in terms of k*

M1

$\therefore \frac{16k}{7} = 8$

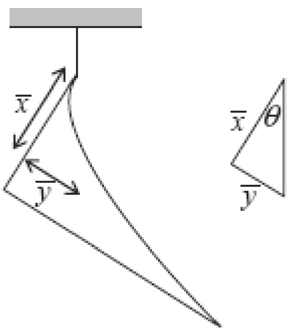
$\therefore k = 3.5$

ft 'their' part (a)

A1F

4

(ii)



$\tan\theta = \frac{\bar{y}}{\bar{x}}$

Use of tan θ

M1

$= \frac{8}{1.6} = 5$

ft 'their' part (b); $\frac{\bar{y}}{\bar{x}}$ *structure*

A1F

$\theta = 78.7°$

A1F

3

[13]

Q3.

(a) $\frac{1}{2}\int_{-2}^{2} y^2 \, dx$

$= \frac{1}{2}\int_{-2}^{2} (16 - 8x^2 + x^4) \, dx$

Attempt to integrate y^2 as a function of x

M1

$= \frac{1}{2}\left[16x - \frac{8x^3}{3} + \frac{x^5}{5}\right]_{-2}^{2}$

Correct integration

A1

$= \frac{256}{15}$

Correct limits applied to their integral

A1F

$\bar{y} = \frac{256/15}{32/3}$

$\bar{y} = \frac{\frac{1}{2}\int_a^b y^2 \, dx}{\text{Area}}$

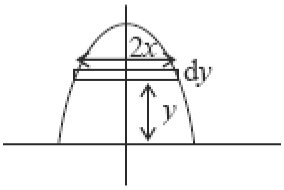
m1

$= \frac{8}{5}$

A1

5

Alternative 1:



$$I = \int_{x=2}^{x=0} 2xy \; dy = \int_2^0 2x(4-x^2)(-2x)\,dx$$

$$= \int_2^0 4x^4 - 16x^2 \; dx$$

Attempt to integrate 2xy as a function of x

(M1)

$$I = \left[\frac{4x^5}{5} - \frac{16x^3}{3}\right]_2^0$$

Correct integration

(A1)

$$I = \frac{256}{15}$$

Correct limit applied to their integral

(A1F)

$$\bar{y} = \frac{I}{32/3} = \frac{8}{5}$$

Their evaluated I ÷ area

(m1)(A1)

Alternative 2:

$$I = \int_{y=0}^{y=4} 2xy \; dy = \int_0^4 2\sqrt{(4-y)}\,y\,dy$$

Attempt to integrate 2xy as a function of y by parts

(M1)

$$I = \left[-\frac{4y}{3}(4-y)^{\frac{3}{2}}\right]_0^4 + \int_0^4 \frac{4}{3}(4-y)^{\frac{3}{2}}\,dy$$

$$I = \left[-\frac{8}{15}(4-y)^{\frac{5}{2}}\right]_0^4$$

Fully integrated

(A1)

$$I = \frac{256}{15}$$

Correct limit applied to their integral

(A1F)

$$\bar{y} = \frac{I}{32/3} = \frac{8}{5}$$

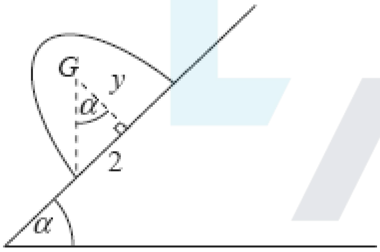
Their evaluated I ÷ area

(m1)(A1)

General for all three versions:
M1 Attempt to integrate an appropriate function of x or y (must apply a full method)
A1 Correct integration
A1F Correct limits applied to their integral
m1 Their evaluated integral ÷ area

A1 Correct answer $\frac{8}{5}$

(b)



$$\tan \alpha = \frac{2}{y}$$

tan α seen

M1

Correct structure – ft error in (a)

A1F

$$\tan \alpha = \frac{2}{8/5} = 1.25$$

Substitute and use of tan⁻¹ – dependent on first M1

m1

$$\alpha = 51°$$

ft error in (a)

A1F

4

[9]

Q4.

(a) (i) Use of I = $\dfrac{\int xy\,\mathrm{d}x}{\int y\,\mathrm{d}x}$

[ρ = 1 allowed throughout]
Stated or used

M1

$\int y\ \mathrm{d}x = \text{semi circle area} = \frac{1}{2}\pi r^2$

A1

Use of diagram / by symmetry, $\bar{x}$ = 21
Explains need for 2 × Integral

A1

$\frac{1}{2}\pi r^2\bar{x} = \int_o^r 2x\sqrt{r^2 - x^2}\,\mathrm{d}x$

Use of $y = \sqrt{r^2 - x^2}$ *and rearrangement*

A1

4

Alternative
Mass of elemental strip $= 2y\,\delta x\,\rho$

[Allow ρ = 1 throughout]
Elemental strip identified

Moment of elemental strip $= 2y\,\delta x\,\rho x$

(M1)

Total moment = $\rho\int_0^r 2xy\,\mathrm{d}x$
Integral formed

(A1)

Use of $\sum(mx) = \left(\sum m\right)\bar{x}$

Gives $\rho\int_0^r 2xy\ \mathrm{d}y = \frac{1}{2}\rho\pi\ r^2\bar{x}$

EXAM PAPERS PRACTICE

Equation formed

(M1)

$$\Rightarrow \int_0^r 2x\sqrt{r^2 - x^2}\,dx = \frac{1}{2}\pi r^2 \bar{x}$$

Use of $\sqrt{r^2 - x^2} = y$ *and* $\frac{1}{2}\pi r^2$

(A1)

(ii) $$\int_0^r 2x\sqrt{r^2 - x^2}\,dx = \left[-\frac{2}{3}\left(r^2 - x^2\right)^{\frac{3}{2}}\right]_0^r$$

Attempt to integrate – inspection or substitution

M1

Integrated correctly and limits substituted – condone sign error

A1

$$= (0) - \left(-\frac{2}{3}r^3\right)$$

$$= \frac{2}{3}r^3$$

$$\therefore \frac{1}{2}\pi r^2 \bar{x} = \frac{2}{3}r^3$$

$$\bar{x} = \frac{4r}{3\pi}$$

Printed answer

A1 **3**

(b) (i) $\frac{2}{3}(1.2) = 0.8\text{ m}$

B1 **1**

(ii) $1.2 + \frac{4(0.5)}{3\pi} = 1.41\text{ m}$

B1 **1**

(iii) Using $\left(\sum m\right)\bar{x} = \sum(mx)$

$$\left[\frac{1}{2}(1)(1.2)\right][0.8]+\left[\frac{1}{2}\pi(0.5)^2\right][1.41]$$

$$=\left[\frac{1}{2}(1)(1.2)+\frac{1}{2}\pi(0.5)^2\right]\bar{x}$$

Form equation – Follow through (b)(i)

M1

One term correct

A1ft

0.48 + 0.553... = 0.992... $\bar{x}$

All terms correct

A1ft

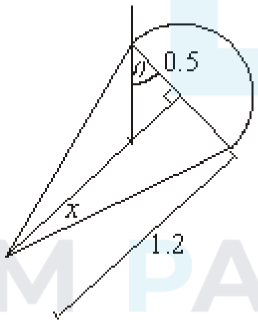
$\therefore$ $\bar{x}$ = 1.04 m

CAO

A1

4

(c)



$$\tan\theta=\frac{(1.2-1.04)}{0.5}=0.32$$

tan q seen

M1

Ratio correct structure = $\frac{(1.2-\bar{x})}{0.5}$ *= 0.32*

A1

$\theta = 18°$

ft ratio error – correct use of tan⁻¹ (ratio)

A1ft

3

[16]

Q5.

(a) $\int_0^a kx\,dx = \left[\frac{kx^2}{2}\right]_0^a = \frac{ka^2}{2}$

Forming integral to find area
Correct area

M1 A1
2

(b) $\frac{ka^2}{2}\bar{x} = \int_0^a kx^2\,dx$

Forming integral to find $\bar{x}$
Correct expression

M1 A1

$\frac{ka^2}{2}\bar{x} = \frac{ka^3}{3}$

$\bar{x} = \frac{2a}{3}$

Evaluating integral and finding $\bar{x}$
Correct $\bar{x}$ *from correct working*

m1 A1
4

(c) $\frac{ka^2}{2}\bar{y} = \int_0^a \frac{k^2x^2}{2}\,dx$

Forming integral to find $\bar{y}$
Correct expression

M1 A1

$\frac{ka^2}{2}\bar{y} = \frac{k^2a^3}{6}$

Evaluating integral and finding $\bar{y}$

M1

$\bar{y} = \frac{ka}{3}$

Correct $\bar{y}$

A1
4

[10]

Q6.

(a) $y = \frac{6}{9}x = \frac{2}{3}x$

Finding equation

M1

Correct equation

A1

2

(b) (i) $\bar{x} = \dfrac{\int_0^9 \frac{2}{3}x^2\,dx}{\frac{1}{2}\times 6\times 9}$

Follow through y = $\frac{3}{2}x$

M1

Forming expression containing appropriate integral

A1

Correct expression

$= \frac{162}{27} = 6$

Evaluation of integral

m1

Correct final answer from correct working

A1

4

(ii) $\bar{y} = \dfrac{\int_0^9 \frac{1}{2}\left(\frac{2x}{3}\right)^2 dx}{27}$

Forming expression containing appropriate integral

M1

Correct expression

A1

$= \frac{54}{27} = 2$

Evaluation of integral

m1

Correct final answer

A1

4

(c) tan $\alpha = \frac{2}{6}$

Use of tan a

M1

Correct expression

A1

$\alpha = 18.4°$

Correct angle

A1

3

[13]

Q7.

(a) $\int_0^4 x^3 \, dx = \left[\frac{x^4}{4}\right]_0^4 = 64$

Integration x^3
cao

M1 A1

2

(b) $64\bar{x} = \int_0^4 x^4 \, dx = \left[\frac{x^5}{5}\right]_0^4$

Integrating x^4

Obtaining $\frac{x^5}{5}$

M1 A1

$64\bar{x} = \frac{1024}{5}$

Finding $\bar{x}$

M1

$\bar{x} = 3.2$

Correct answer from correct working

A1

4

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(c) $64\bar{y} = \int_0^4 \frac{1}{2}x^6\,dx = \left[\frac{x^7}{14}\right]_0^4$

Integrating $\frac{1}{2}x^6$

Obtaining $\frac{x^7}{14}$

M1

$64\bar{y} = \frac{16384}{14}$

Finding $\bar{y}$

M1

$\bar{y} = 18.3$

Correct answer (awrt 18.3)

A1

4

(d) $\tan\theta = \frac{4 - 3.2}{18.286}$

Using tan with a fraction
Numerator

M1 A1

$\theta = 2.51°$

ft Denominator
ft Angle

A1ft A1ft

4

[14]

Examiner reports

Q1.

This question was very well attempted. However students should be aware that when the request in a question is 'Show that…' full working needs to be shown with clear indication of how any formulae are being used. In the best solutions students began by stating that $\bar{x} = \dfrac{\int xy\,dx}{\int y\,dx}$ before evaluating the numerator and substituting 'A' for the denominator. Only a handful of students incorrectly attempted to use formulae appropriate to three-dimensional problems, a significant improvement on the past.

Q2.

Candidates showed improved knowledge of appropriate formulae in this question. Almost all candidates correctly obtained $4k$ in part (a), although a few candidates quoted the correct integral but did not evaluate it. The formula in part (b) was universally correctly used by candidates who quoted it. The formula in part (c) was used less successfully, with the $\frac{1}{2}$ going astray at times. There were a number of candidates who tried to obtain the solution from first principles; this inevitably led to several types of error. Part (c)(ii) was excellently answered, the technique being well remembered from Mechanics 2.

Q3.

It was disappointing to see a significant number of candidates make heavy weather of this question. Candidates should be aware of the formulae stated in the formulae booklet. A number of candidates started from first principles and needed to do a lot of work to score the first mark. Many who chose this approach could not complete the integration, particularly if they had left the integrand as a function of y. Numerous errors occurred with incorrect limits. Part (b) was done well and added to candidates' scores.

Q4.

Candidate performed better than expected here. In part (a)(i) two methods were credited. The standard 'book' method using elemental strips was rarely seen, although this is a useful skill for candidates to develop. Candidates who quoted the standard formula could gain full credit but the final A1 was for explaining from where the '2' had originated. Part (a)(ii) proved more problematic – the integration being beyond many candidates. The best solutions used inspection of substitution with a variety of successful substitutions used including use of trigonometry. Many candidates scored full marks on part (b)(i) and part (b)(ii) although not all candidates were aware of the significance of part a) or the standard result for a triangle. Careless errors were seen in part (b)(iii) with the area of the semicircle being stated as $\pi/2$ or $\pi/4$. The best solutions started with a table giving clear values for areas and distances. Almost all of the candidates scored either 2 or 3 marks in part (c) and this is clearly well understood by candidates.

Q5.

While most candidates had very few problems finding the area of the lamina, parts (b) and (c) caused more difficulty for some candidates. In general, candidates either seemed to know how to tackle this question or made very little progress. A few used techniques that were not correct for a lamina. Of those who realised how to approach the problem, the vast majority produced correct solutions. A few did make algebraic errors in their working.

Q6.

There were some good responses to this question. Virtually all candidates found the equation of the line without difficulty. A good number of candidates were able to obtain the value of 6 by a correct method in part (b)(i), but part (b)(ii) was found to be a little more difficult. Several candidates used incorrect formula, such as one that would have been appropriate for a volume of revolution. Some were a factor of 2 out in part (b)(ii), often due to omitting the factor of 1/2 in their integral, and a few also used the wrong limits, using the maximum value of y instead of x. Some candidates who made a variety of minor errors when integrating. Part (c) was generally done very well.

Q7.

The candidates did very well with this question and many scored full marks. Parts (a) and (b) were done particularly well. In part (c), the candidates who could form the correct integral generally found the y coordinate correctly. Some candidates did find forming the correct integral difficult. Part (d) was done well, showing that many candidates understood the principles involved.